23 Jan 2026 Shift 2 (Actual paper)

Math Sec 1

1. If the mean and the variance of the data

Class	4-8	8-12	12-16	16-20
Frequency	3	λ	4	7

are μ and 19 respectively, then the value of $\lambda + \mu$ is

(A) 19 (B) 20 (C) 18 (D) 21

2. The least value of $\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2$ is

(A) $4+\sqrt{10}$ (B) -1

(C) $4-\sqrt{10}$ (D) 1

3. *The system of linear equations*

$x + y + z = 6$

$2x + 5y + az = 36$

$x + 2y + 3z = b$

has

(A) unique solution for $a = 8$ and $b = 16$

(B) infinitely many solutions for $a = 8$ and $b = 16$

(C) unique solution for $a = 8$ and $b = 14$

(D) infinitely many solutions for $a = 8$ and $b = 14$

4. An equilateral triangle OAB is inscribed in the parabola $y^2 = 4x$ with the vertex O at the vertex of the parabola. Then the minimum distance of the circle having AB as a diameter from the origin is

(A) $2(8-3\sqrt{3})$ (B) $4(3-\sqrt{3})$

(C) $4(6+\sqrt{3})$ (D) $2(3+\sqrt{3})$

5. Let $A = \{0, 1, 2, \ldots, 9\}$. Let R be a relation on A defined by $(x, y) \in \mathrm{R}$ if and only if $|x-y|$ is a multiple of 3.
Given below are two statements:
Statement I: $n(\mathrm{R}) = 36$.
Statement II: R is an equivalence relation.
In the light of the above statements, choose the correct answer from the options given below.

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

6. Let $\vec{a} = \hat{\imath} - 2\hat{\jmath} + 3\hat{k}$, $\vec{b} = 2\hat{\imath} + \hat{\jmath} - \hat{k}$, $\vec{c} = \lambda\hat{\imath} + \hat{\jmath} + \hat{k}$ and $\vec{v} = \vec{a} \times \vec{b}$. If $\vec{v}\cdot\vec{c} = 11$ and the length of the projection of $\vec{b}$ on $\vec{c}$ is p, then $9p^2$ is equal to

(A) 6 (B) 4 (C) 9 (D) 12

7. Let $I(x) = \int \frac{3\,dx}{(4x+6)\left(\sqrt{4x^2+8x+3}\right)}$ and $I(0) = \frac{\sqrt{3}}{4} + 20$. $I\left(\frac{1}{2}\right) = \frac{a\sqrt{2}}{b} + c$, where $a, b, c \in \mathrm{N}$, $\gcd(a, b) = 1$, then $a + b + c$ is equal to

(A) 31 (B) 28 (C) 29 (D) 30

8. The sum of all the real solutions of the equatic $\log_{(x+3)}(6x^2 + 28x + 30) = 5 - 2\log_{(6x+10)}(x^2 + 6x + 9$ is equal to

(A) 0 (B) 2 (C) 1 (D) 4

9. Let PQ be a chord of the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$, perpe dicular to the x-axis such that OPQ is an equilateral tria gle, O being the centre of the hyperbola. If the eccentrici of the hyperbola is $\sqrt{3}$, then the area of the triangle OPQ

(A) $2\sqrt{3}$ (B) $\frac{11}{5}$

(C) $\frac{9}{5}$ (D) $\frac{8\sqrt{3}}{5}$

10. Let $\mathrm{A}(1, 2)$ and $\mathrm{C}(-3, -6)$ be two diagonally opposi vertices of a rhombus, whose sides AD and BC are para lel to the line $7x - y = 14$. If $\mathrm{B}(\alpha, \beta)$ and $\mathrm{D}(\gamma, \delta)$ are the other two vertices, then $|\alpha + \beta + \gamma + \delta|$ equal to

(A) 6 (B) 9 (C) 1 (D) 3

11. Bag A contains 9 white and 8 black balls, while bag contains 6 white and 4 black balls. One ball is random picked up from the bag B and mixed up with the balls the bag A. Then a ball is randomly drawn from the bag If the probability that the ball drawn is white is $\frac{p}{q}$, whe $\gcd(p, q) = 1$, then what is the value of $p + q$?

(A) 21 (B) 22 (C) 24 (D) 23

12. Consider two sets $\mathrm{A} = \{x \in \mathrm{Z} : |(|x-3| - 3)| \le 1\}$ ar $\mathrm{B} = \{x \in \mathrm{R} - \{1, 2\} : \frac{(x-2)(x-4)}{x-1}\log_e(|x-2|) = 0\}$.
Then the number of onto functions $f : \mathrm{A} \to \mathrm{B}$ is equal t

(A) 81 (B) 32 (C) 79 (D) 62

13. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} \times \vec{b} = 2(\vec{a}$ $\vec{c})$. If $|\vec{a}| = 1$, $|\vec{b}| = 4$, $|\vec{c}| = 2$, and the angle betwe $\vec{b}$ and $\vec{c}$ is 60°, then $|\vec{a}\cdot\vec{c}|$ is equal to

(A) 0 (B) 2 (C) 1 (D) 4

14. If the points of intersection of the ellipses $x^2 + 2y^2$ $6x - 12y + 23 = 0$ and $4x^2 + 2y^2 - 20x - 12y + 35 =$ lie on a circle of radius r and centre (a, b), then the valu of $ab + 18r^2$ is

(A) 55 (B) 53 (C) 51 (D) 52

15. If

$$f(x)=\begin{cases}\frac{a|x|+x^2-2(\sin|x|)(\cos|x|)}{x} & , x\neq 0\\ b & , x=0\end{cases}$$

is continuous at $x=0$, then $a+b$ is equal to

(A) 4 (B) 1 (C) 0 (D) 2

16. Let $\sum_{k=1}^{n} a_k=\alpha n^2+\beta n$. If $a_{10}=59$ and $a_6=7a_1$, then $\alpha+\beta$ is equal to

(A) 7 (B) 12 (C) 3 (D) 5

17. The area of the region enclosed between the circles $x^2+y^2=4$ and $x^2+(y-2)^2=4$ is:

(A) $\frac{2}{3}(4\pi-3\sqrt{3})$ (B) $\frac{2}{3}(2\pi-3\sqrt{3})$

(C) $\frac{4}{3}(2\pi-\sqrt{3})$ (D) $\frac{4}{3}(2\pi-3\sqrt{3})$

18. Let $\frac{\pi}{2}<\theta<\pi$ and $\cot\theta=-\frac{1}{2\sqrt{2}}$. Then the value of

$\sin\left(\frac{15\theta}{2}\right)(\cos 8\theta+\sin 8\theta)+\cos\left(\frac{15\theta}{2}\right)(\cos 8\theta-\sin 8$

is equal to

(A) $-\frac{\sqrt{2}}{\sqrt{3}}$ (B) $\frac{\sqrt{2}-1}{\sqrt{3}}$

(C) $\frac{1-\sqrt{2}}{\sqrt{3}}$ (D) $\frac{\sqrt{2}}{\sqrt{3}}$

19. The number of ways, in which 16 oranges can be distril uted to four children such that each child gets at least o orange, is

(A) 384 (B) 403 (C) 429 (D) 455

20. If $z=\frac{\sqrt{3}}{2}+\frac{i}{2}$, $i=\sqrt{-1}$, then $\left(z^{201}-i\right)^8$ is equal to

(A) 256 (B) 1 (C) 0 (D) -1

Math Sec 2

21. Let S denote the set of 4-digit numbers $abcd$ such that $a>b>c>d$ and P denote the set of 5-digit numbers having product of its digits equal to 20. Then $n(S)+n(P)$ is equal to _____

22. The number of elements in the set $S=\{x: x\in[0,100]$ and $\int_0^x t^2\sin(x-t)dt=x^2\}$ is ______

23. If the image of the point $P(a,2,a)$ in the line $\frac{x}{2}=\frac{y+a}{1}=\frac{z}{1}$ is Q and the image of Q in the line $\frac{x-2b}{2}=\frac{y-a}{1}=\frac{z+2b}{-5}$ is P, then $a+b$ is equal to ____ .

24. Let $A=\begin{bmatrix}0 & 2 & -3\\ -2 & 0 & 1\\ 3 & -1 & 0\end{bmatrix}$ and B be a matrix such th $B(I-A)=I+A$. Then the sum of the diagonal el ments of B^TB is equal to ____

25. If the solution curve $y=f(x)$ of the differential equatic $(x^2-4)y'-2xy+2x(4-x^2)^2=0,\ x>2,$ pass through the point $(3,15)$, then the local maximum valu of f is ____

Phy Sec 1

26. A block is sliding down on an inclined plane of slope θ and at an instant $t=0$ this block is given an upward momentum so that it starts moving up on the inclined surface with velocity u. The distance (S) travelled by the block before its velocity become zero, is ____ .
($g=$ gravitational acceleration)

(A) $\frac{2u^2}{g\cos\theta}$

(B) $\frac{u^2}{4g\sin\theta}$

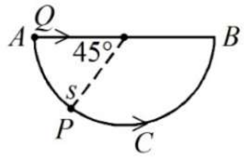
(C) $\frac{u^2}{2g\cos\theta}$

(D) $\frac{u^2}{\sqrt{2}g\cos\theta}$

27. One mole of an ideal diatomic gas expands from volume V to $2V$ isothermally at a temperature $27°C$ and does W joule of work. If the gas undergoes same magnitude of expansion adiabatically from $27°C$ doing the same amount of work W, then its final temperature will be (close to) ____ $°C$.
$(\log_e 2=0.693)$

(A) -56 (B) -189 (C) -117 (D) -30

28. A bead P sliding on a frictionless semi-circular strir (ACB) and it is at point S at $t=0$ and at this instant tl horizontal component of its velocity is v. Another bead of the same mass as P is ejected from point A at $t=$ along the horizontal string AB, with the speed v, frictic between the beads and the respective strings may be n glected in both cases. Let t_P and t_Q be the respectiv times taken by beads P and Q to reach the point B, the the relation between t_P and t_Q is



(A) $t_P>1.25t_Q$ (B) $t_P<t_Q$

(C) $t_P>t_Q$ (D) $t_P=t_Q$